

TOYOOKA, Japan

WMO: 477470

Lat: 35.5358N

Lon: 134.8222E

Elev: 6

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.2	(c) -1.3	(d) -5.9	(e) 2.3	(f) 3.7	(g) -4.6	(h) 2.6	(i) 4.0	(j) 6.2	(k) 4.4	(l) 5.4	(m) 4.4	(n) 1.4	(o) 130	(p) 0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.0	(c) 34.8	(d) 25.6	(e) 33.6	(f) 25.3	(g) 32.2	(h) 24.9	(i) 26.5	(j) 32.2	(k) 26.0	(l) 31.4	(m) 25.5	(n) 30.5	(o) 3.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.2	28.6	24.6	19.6	28.3	24.1	19.0	28.0	82.7	32.4	80.4	31.5	78.3	30.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	5.2	4.4	-4.2	37.3	1.2	0.9	-5.1	37.9	-5.8	38.4	-6.5	38.9	-7.4	39.5
			-4.6	27.5	1.3	0.7	-5.6	28.0	-6.4	28.4	-7.1	28.8	-8.1	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.6	4.1	7.5	13.0	18.3	22.3	26.7	27.9	23.3	17.4	11.7	6.4		
	DBStd	8.81	2.25	2.76	3.31	3.57	2.81	2.57	2.53	2.24	3.01	2.90	3.00	2.92		
	HDD10.0	602	198	167	93	12	0	0	0	0	0	0	16	117		
	HDD18.3	2013	456	398	335	163	35	2	0	0	1	52	201	371		
	CDD10.0	2521	0	2	16	102	259	370	518	554	399	231	66	5		
	CDD18.3	890	0	0	0	3	36	122	259	296	151	24	1	0		
	CDH23.3	7628	0	0	1	47	305	751	2392	3037	993	102	2	0		
	CDH26.7	2965	0	0	0	7	62	206	998	1359	317	15	0	0		
(14) Wind	WSAvg	1.7	1.4	1.6	1.8	2.0	2.0	1.7	1.8	1.8	1.6	1.4	1.4	1.5		
(15) Precipitation	PrecAvg	2020	238	177	143	99	133	149	172	142	236	162	159	208		
	PrecMax	2511	384	281	214	164	436	285	429	336	517	489	350	432		
	PrecMin	1484	108	89	57	42	37	76	49	13	66	50	68	98		
	PrecStd	212	67	51	45	38	77	54	90	80	114	88	66	77		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	17.9	22.0	27.5	30.4	32.8	36.0	36.6	34.2	28.6	22.8	17.3		
		MCWB	9.0	12.7	13.2	16.4	19.1	22.7	25.7	25.7	24.9	21.7	16.2	12.7		
	2%	DB	10.9	13.8	18.8	24.4	28.4	30.6	34.3	35.3	32.1	25.9	20.3	14.7		
		MCWB	7.2	9.4	12.1	15.0	18.0	22.4	25.5	25.7	24.3	19.9	15.4	10.7		
	5%	DB	9.2	11.1	16.5	22.3	26.7	29.0	33.1	34.2	30.1	24.1	18.5	12.8		
		MCWB	6.4	7.7	11.1	14.4	17.5	21.6	25.3	25.5	23.7	18.9	14.6	9.5		
	10%	DB	7.5	9.0	14.2	20.2	24.8	27.5	31.8	32.9	28.3	22.5	16.8	11.2		
		MCWB	5.2	6.1	9.9	13.6	17.0	21.2	25.0	25.3	22.9	17.9	13.4	8.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	14.1	15.1	19.2	21.9	25.0	27.1	27.4	26.0	23.2	18.2	14.0		
		MCDB	12.6	17.7	18.4	22.7	26.0	30.0	33.0	33.7	31.1	27.0	20.3	14.9		
	2%	WB	8.0	10.1	13.4	17.0	20.3	24.1	26.3	26.7	25.1	21.4	16.6	11.7		
		MCDB	10.1	12.8	16.9	21.9	24.4	28.0	32.0	32.5	30.0	24.4	19.0	13.6		
	5%	WB	6.5	8.1	11.7	15.7	19.1	23.2	25.8	26.2	24.4	20.0	15.4	10.0		
		MCDB	8.7	10.6	15.6	20.3	23.8	26.9	31.3	31.8	28.6	22.5	17.6	12.3		
	10%	WB	5.4	6.3	10.1	14.6	18.1	22.3	25.3	25.7	23.7	18.8	14.1	8.7		
		MCDB	7.3	8.7	13.8	18.8	23.0	26.0	30.4	30.9	27.3	21.4	16.4	10.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	7.5	9.9	11.6	11.1	8.8	8.2	9.0	8.1	8.9	8.5	7.3		
		MCDBR	9.9	12.2	15.0	15.8	15.1	11.8	10.7	11.0	10.4	10.8	11.2	10.4		
	5% WB	MCWBR	6.3	7.9	8.7	7.6	6.2	4.6	3.5	3.4	3.9	5.1	6.3	6.4		
		MCWBR	9.1	11.2	12.3	11.8	10.9	8.6	9.6	9.6	8.8	8.4	9.3	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.432	0.466	0.530	0.552	0.553	0.575	0.575	0.544	0.494	0.455	0.434	0.426		
		taud	2.025	1.961	1.832	1.838	1.895	1.915	1.973	2.054	2.149	2.157	2.117	2.061		
	Ebn at Noon		725	752	739	750	756	738	734	749	766	758	723	702		
		Edh at Noon	133	159	197	206	197	193	181	164	142	129	121	122		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.40	3.50	4.62	5.13	4.76	4.76	4.94	3.82	3.07	2.12	1.58		
	RadStd		0.14	0.27	0.32	0.41	0.51	0.38	0.72	0.56	0.35	0.28	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page